

Fernando Velilla Hurtado

Turin, Italy fernandovelilla1@gmail.com +39 344 484 2414 fvelillah.github.io/portfolio/ in fernandovelillahurtado fvelillah

EDUCATION

MSc	Politecnico di Torino - Turin, Italy , Computer Engineering	2025 – 2027
	- Current GPA: 27/30 - Track in Automation & Intelligent Cyber-Physical Systems Coursework: Computer Architectures, System and Device Programming, Data Science and Database Technology, ML for Vision, Cyber-Physical Systems	
BS	EIA University - Medellín, Colombia , Mechatronics Engineering	2021 – 2025
	- GPA: 4.5/5 Coursework: Computer-Aided Design for Mechanics, Analog and Digital Electronics, Robotics, Automatic Control	

PROJECTS

Multi-Agent AI Framework for Industrial Fleet Management <i>Team of 3, Politecnico di Torino</i>	github/arol-project
- Built a full-stack LLM-based troubleshooting assistant for AROL (manufacturer of capping machines based in Italy) using React/TypeScript frontend, Django/PostgreSQL backend, Dockerized (3 services). - Implemented a LangGraph multi-agent system with swappable local/cloud LLMs and a Qdrant RAG pipeline over +1,200 manual pages. - Answered over 90% benchmark questions correctly, each requiring reasoning over telemetry, alarms, tickets, contracts, manuals and other sources of data.	
Unsupervised Anomaly Detection with PyTorch <i>Team of 3, Politecnico di Torino</i>	github/anomaly
- Built and compared two convolutional autoencoders for unsupervised defect detection on MVTec AD (capsule), reimplementing a self-supervised model with stacked dilated convolutions and synthetic defect augmentation from a 2024 paper. - Raised pixel-level ROC-AUC from 0.74 to 0.84 over a residual-autoencoder baseline.	
Autonomous vehicle with ROS2	research-report
- Developed a navigation stack in ROS2 and low-level firmware in Micro-ROS for a differential drive robot within the ASIMOV research group. - Validated waypoint trajectory planning in Webots before hardware deployment.	
SCARA robotic arm	project-website
Developed a 3-DOF (RR-P) robotic arm for Industrial Robotics course, implementing forward/inverse kinematics and end-effector trajectory planning.	

EXPERIENCE

Robot.com , Maintenance Analyst	Evansville, IN Dec 2024 – May 2025
- As the sole technical analyst on the field, diagnosed and repaired hardware, electromechanical, and firmware faults across a fleet of autonomous delivery robots on a university campus, growing operational robots from 20 to 33 (+65%) to meet rising demand. - Debugged robot firmware and field issues, and implemented electronics board flashing procedures. - Deployed and electrically validated 5 wireless charging stations on campus, supporting 20 robots.	
Vidycom S.A.S , Automation and Controls Engineer Intern	Medellín, Colombia Nov 2023 – Feb 2024
- Simulated an air purification system in Automation Studio to validate PLC logic and firmware before hardware commissioning, catching over 10 logical errors. - Improved observability by deploying an interactive Grafana telemetry dashboard on InfluxDB, ingesting real-time sensor data over MQTT to track system health and detect operational anomalies.	

SKILLS

Languages: Python, C, C++, SQL, TypeScript
Frameworks/ML: PyTorch, CUDA, LangGraph, ROS2, React, Django
Tools & Infra: Git, Linux, PostgreSQL, Qdrant, Grafana, Docker, InfluxDB
Other: MATLAB/Simulink, FreeRTOS, LabVIEW, Automation Studio, Autodesk Inventor
Spoken languages: Spanish (Native), English (IELTS Score: 7.5), Italian (B1)